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Aidi Ahmi & Louis Smidt

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AI-driven auditing: trends, themes, and research trajectories

Aidi Ahmi ^a and Louis Smidt ^b

^aAccounting Information Systems Research and Development Institute (AISRED), Tunku Puteri Intan Safinaz School of Accountancy, Universiti Utara Malaysia 06010 UUM Sintok, Sintok, Malaysia; ^bDepartment of Auditing, Faculty of Economics and Finance, Tshwane University of Technology, Pretoria, South Africa

ABSTRACT

This study examines the evolution of artificial intelligence (AI) in auditing by systematically synthesizing 26 years of global scholarship to address conceptual fragmentation in the field. Guided by the PRISMA protocol, it analyses 269 Scopus-indexed journal articles, harmonized using OpenRefine and biblioMagika, and examined through performance analysis, co-occurrence mapping, and life-cycle modeling with biblioMagika, VOSviewer, and Biblioshiny. Integrating behavioral, technological, and governance perspectives, the study clarifies the domain's intellectual foundations, thematic structure, and temporal development. Findings show rapid expansion since 2019, reflecting a shift from early rule-based and expert systems toward advanced machine learning, explainable AI, and generative AI applications. Six thematic clusters organize the literature: computational audit analytics; governance and responsible AI; AI-audit integration; blockchain-enabled audit evidence; automation and audit quality; and analytics-driven continuous auditing. Temporal evidence indicates an increasing normative orientation, with stronger emphasis on transparency, accountability, trust, and ethical governance. Life-cycle modeling suggests the field remains in a steep growth phase, indicating substantial scope for further theoretical and practical advancement. Although limited to Scopus-indexed journal articles, the results offer a foundation for research on auditor–AI interaction, behavioral effects of automation, governance mechanisms for AI assurance, and institutional variation in technology adoption. By providing a longitudinal science-mapping analysis, the study consolidates publication trends, intellectual structure, thematic evolution, and life-cycle forecasting, and identifies research opportunities likely to shape audit methodology, governance, and professional judgment in an era of intelligent systems.

1. Introduction

Artificial intelligence (AI) is transforming the audit profession at an unprecedented pace. Audit firms are increasingly integrating machine learning, robotic process automation, natural language processing, and generative AI into core audit workflows to support fraud detection, risk assessment, and evidence evaluation (Fedyk et al., 2022; Huang & Vasarhelyi, 2019; Issa et al., 2016). These tools go well beyond automating routine tasks; they can analyze unstructured data, summarize complex narratives, extract key audit matters, and deliver real-time insights that push the boundaries of human judgment. Simultaneously, recent studies point to unexpected risks and behavioral concerns. Auditors might over-rely on AI outputs or neglect professional skepticism when dealing with opaque models (Brown-Liburd et al., 2015), while ethical issues, accountability challenges, and governance pressures related to AI adoption continue to grow (Lehner et al., 2022; Munoko et al., 2020). As regulators call on the profession to understand how digital tools influence audit quality, it is clear that AI is no longer optional; it is essential to the future of auditing.

Yet the scholarly landscape on AI in auditing has grown unevenly and rapidly, producing a complex body of work spanning accounting information systems, behavioral auditing, machine learning, ethics, and computer science. Foundational research on expert systems, continuous auditing, and automated evidence collection continues to influence current discourse (Rezaee et al., 2002), even as newer studies explore generative AI, explainable AI, and data governance. Despite this proliferation, the field lacks a consolidated understanding of how AI-related audit research has evolved over the last two decades, which themes have shaped its intellectual structure,

and where future research is required. This absence of synthesis is problematic at a time when audit firms and regulators are making high-stakes decisions about AI integration, governance, and assurance responsibilities.

Although research on artificial intelligence in auditing has expanded rapidly, the cumulative development of this literature remains difficult to assess. The speed and diversity of recent technological advances, ranging from machine learning and automation to generative and privacy-preserving AI, have produced a growing but fragmented body of scholarship. As a result, it is increasingly challenging for scholars, practitioners, and regulators to discern how the field has evolved, which research streams have matured, and where meaningful future inquiry is most needed. Without a systematic perspective, there is a risk that academic research will respond reactively to isolated innovations rather than contribute proactively to the theoretical and practical shaping of intelligent auditing.

Addressing this challenge requires a comprehensive bibliometric and science-mapping approach that situates individual contributions within their broader intellectual and temporal context. Accordingly, this study systematically maps and analyses the global development of AI-in-auditing scholarship over a 26-year period, examining its publication dynamics, intellectual structure, thematic evolution, and emerging research frontiers. In doing so, the study provides an integrated foundation for understanding how the field has developed and where it is heading. To achieve this aim, the study addressed the following research questions:

RQ1: How has research on artificial intelligence in auditing evolved in terms of publication output, citation impact, and overall growth?

RQ2: Which journals and documents have exerted the most significant influence on the development of AI-in-auditing research?

RQ3: What conceptual and thematic structures characterize the field, and how are these themes interconnected?

RQ4: How have thematic emphases, methodological approaches, and technological focuses evolved across time?

RQ5: What research gaps and opportunities emerge from the field's intellectual and thematic evolution, and how can they guide future inquiry into AI-enabled auditing?

The remainder of the paper is structured as follows: [Section 2](#) presents the literature review; [Section 3](#) outlines the methodology; [Section 4](#) reports the results of the bibliometric analyses; [Section 5](#) discusses the findings, theoretical and practical implications, and future research directions; and [Section 6](#) concludes the study.

2. Literature review

Research on automation and AI in auditing has advanced through several conceptual phases, shaped by technological, organizational, and audit-methodological developments. Early work focused on building deterministic expert systems capable of encoding auditor knowledge and producing consistent decisions. Wongpinunwatana et al. (2000) demonstrated that such systems could enhance novice auditors' judgment accuracy by reducing cognitive variability, while Rezaee et al. (2002) conceptualized continuous auditing as an automated, rule-based process for real-time exception testing. These foundational contributions reflected an era in which audit data were predominantly structured and computational models relied on codified rules. As organizations adopted enterprise systems and generated larger volumes of digital records, researchers began to explore data-driven approaches. Koskivaara (2004) showed that statistical classifiers and early neural networks outperform traditional methods in fraud detection tasks, signaling a methodological shift toward learning-based, rather than rule-based, audit technologies.

By the mid-2010s, scholarship increasingly recognized that technological change was transforming not only audit techniques but also professional roles, firm structures, and client interactions. Issa et al. (2016) proposed a research agenda positioning AI as a supplement to human judgment, emphasizing the need to formalize audit processes and redesign workforce competencies. Brown-Liburd et al. (2015) further highlighted that big data analytics fundamentally reshape how auditors frame risk, evaluate evidence, and allocate attention, while also introducing behavioral risks such as automation bias. These insights align with Vasarhelyi and Romero's (2014) argument that continuous auditing and real-time data availability alter the temporal nature of assurance, shifting audits from periodic to ongoing processes. Taken together, these works reposition AI not merely as a technical tool but as a driver of structural and cognitive change within the audit function.

The rise of automation and AI also triggered substantial changes in audit workflows. Kokina and Davenport (2017) documented how AI-enabled tools redistribute labor across audit teams, automate pattern recognition tasks, and reshape interactions with clients. Huang and Vasarhelyi (2019) formalized the integration of robotic process automation (RPA) into routine audit procedures, demonstrating how automation enhances efficiency and frees auditors to focus on higher-order analytical tasks. Fedyk et al. (2022) provided large-scale archival evidence indicating that investments in AI improve audit quality by reducing restatement risk, while simultaneously lowering fees and reshaping labor structures, suggesting that AI adoption has economic and organizational consequences beyond technical performance.

More recent scholarship highlights that the adoption of digital technologies in auditing unfolds within broader institutional and organizational contexts rather than being driven solely by technological capability. Sharma et al. (2022) show that the COVID-19 pandemic acted as a catalyst for the rapid uptake of remote auditing tools, as firms, tiny and resource-constrained practices, faced normative pressures to maintain audit continuity despite restricted physical access to clients. Their findings illustrate how external shocks can accelerate digital transformation in auditing, revealing that environmental demands, professional expectations, and the practical challenges of evidence collection in a crisis setting shape technological adoption. In parallel, Rahman et al. (2024) provide large-scale empirical evidence on how AI adoption materially alters audit outcomes, demonstrating that firms utilizing AI technologies exhibit changes in audit effort, reduced reporting lags, and lower restatement risk. Their results suggest that AI integration influences audit quality and efficiency, underscoring that the implications of digital transformation extend beyond process adjustments to measurable performance effects.

Despite the growing volume of research on artificial intelligence in auditing, bibliometric studies that seek to synthesize this literature remain scarce and methodologically constrained. Recent contributions demonstrate increasing scholarly attention but provide only fragmented and partial insights. For example, Badea Florea et al. (2025) mapped AI research across business disciplines but did not isolate the audit domain or examine its internal intellectual structure. Elena (2025) similarly documented AI adoption trends using quantitative indicators, yet offered limited conceptual integration specific to auditing. Flayyih et al. (2024) surveyed emerging technologies in finance, while Jeyaram and Suhaimi (2025) focused narrowly on AI and audit quality; both relied on restricted datasets and short observation windows, limiting their ability to capture long-term knowledge development. More recently, Thottoli (2024) combined a small-scale bibliometric exercise with survey-based analysis to examine ICT and AI adoption in auditing practice; however, the bibliometric component was confined to only three Scopus-indexed articles published between 2019 and 2020, precluding meaningful longitudinal analysis or identification of thematic evolution and intellectual foundations. Collectively, these studies are characterized by limited temporal depth, narrow samples, descriptive emphasis, and minimal use of advanced science-mapping techniques capable of revealing conceptual clusters, knowledge trajectories, and research frontiers.

3. Methods

This study applied a bibliometric research design to systematically analyze the scientific landscape related to AI in auditing (see Figure 1). Bibliometric methods provide a structured, transparent, and reproducible means of identifying influential publications, mapping research themes, and tracing intellectual evolution (Donthu et al., 2021). The methodological procedures adhered to the modified Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Ahmi et al., 2025; Moher et al., 2009) and

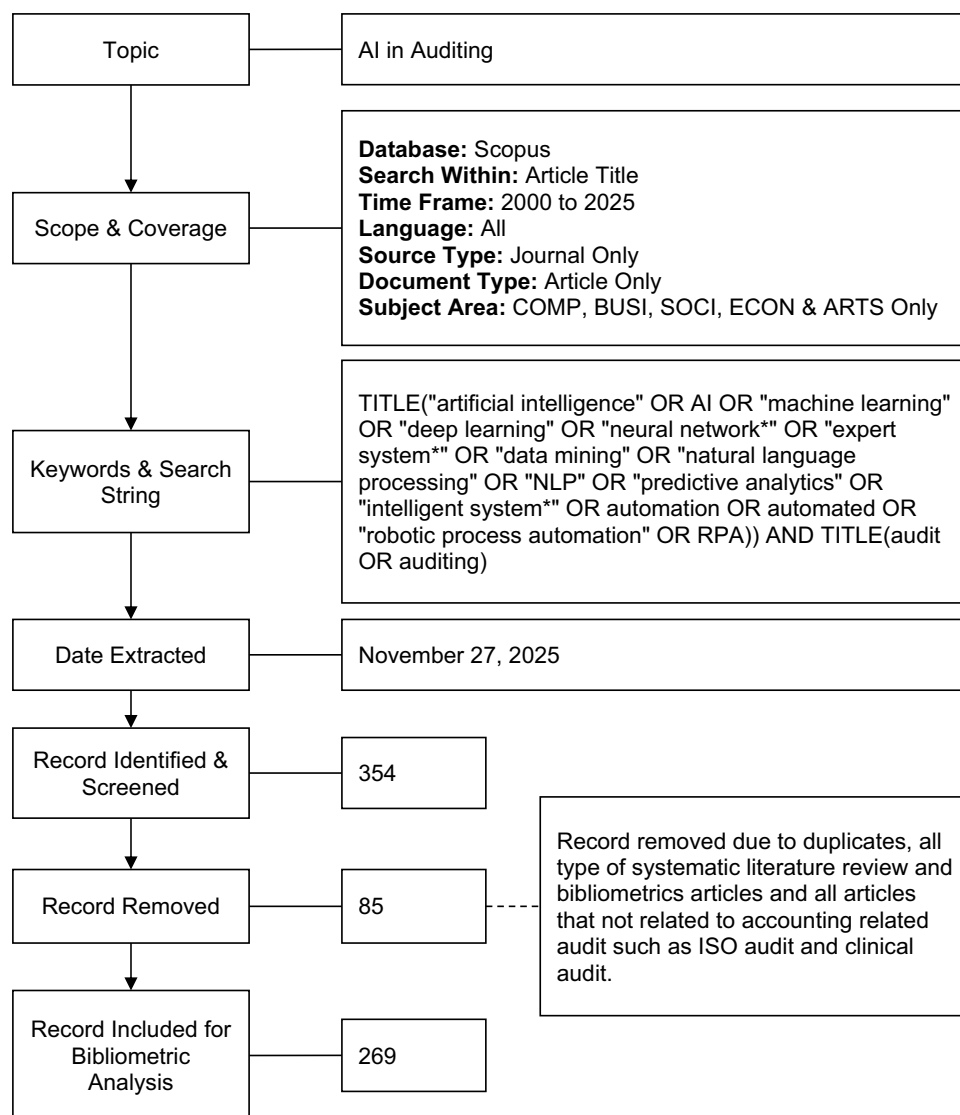


Figure 1. Flow diagram of the search strategy. Source: Ahmi et al. (2025), Moher et al. (2009).

followed established protocols in bibliometric science mapping (Aria & Cuccurullo, 2017; Zupic & Čater, 2015). The stages of searching, screening, data cleaning, and analysis were implemented to ensure methodological rigor, conceptual precision, and reproducibility.

The Scopus database served as the sole data source due to its extensive journal coverage, multidisciplinary indexing depth, and robust metadata architecture. Scopus indexes approximately 24,000 active journals across the scientific, technical, social science, and humanities fields, surpassing the breadth of coverage of the Web of Science Core Collection (Mongeon & Paul-Hus, 2016). Prior studies have demonstrated that Scopus provides superior representation in business, economics, social sciences, and computer science (Burnham, 2006; Chadegani et al., 2013), making it particularly suitable for research at the intersection of AI and auditing. Furthermore, Scopus offers richer citation data and more comprehensive author affiliations than Web of Science, enhancing the accuracy of co-authorship, institutional, and country-level analyses (Harzing & Alakangas, 2016). Given these strengths, Scopus is widely recognized as the most appropriate database for bibliometric analyses in interdisciplinary research domains such as technology-enhanced auditing (Falagas et al., 2008).

The search strategy relied on a title-specific query to ensure conceptual precision and minimize the retrieval of irrelevant documents. Title-based searches are recommended when the research objective requires strong topical relevance because authors reserve the title for the most central concepts of their

work (Zupic & Čater, 2015). Broader fields, such as abstracts and keywords, often introduce conceptual noise, particularly for terms like “audit,” which may refer to clinical, environmental, quality assurance, or other unrelated audit concepts. Restricting the search to titles ensured that the resulting corpus represented studies whose primary focus explicitly aligned with AI and auditing rather than peripheral or ambiguous uses of the terms.

The search was constrained to journal articles, written in any language, published between 2000 and 2025, and indexed in the subject areas of Computer Science, Business, Social Sciences, Economics, and Arts & Humanities. These restrictions reflect standard bibliometric practice, where peer-reviewed journal articles constitute the most suitable unit of scientific analysis (Donthu et al., 2021). At the same time, the date range captured the emergence and maturation of modern AI methods, including machine learning, deep learning, NLP, and robotic process automation (Kokina & Davenport, 2017). The search was executed in Scopus on 27 November 2025, yielding an initial dataset of 354 documents.

A systematic screening process, following the modified PRISMA guidelines, was conducted to eliminate duplicate and irrelevant publications. Excluded items comprised clinical audits, ISO audits, environmental audits, nursing audits, and other non-accounting uses of the term “audit,” as these do not pertain to external, internal, or technology-enabled auditing within the accounting domain. Systematic literature reviews, bibliometric studies, editorials, conference abstracts, and non-empirical articles were also removed to ensure that the dataset consisted solely of original, peer-reviewed research articles. After exclusions, 269 articles remained for bibliometric analysis, representing the core body of knowledge on AI in the auditing discipline.

3.1. Tools, data cleaning and harmonization

The initial cleaning stage involved standardizing authors’ keywords using OpenRefine (Ahmi, 2023), a widely adopted tool for text transformation, clustering, and reconciliation. OpenRefine’s clustering algorithms (including key collision and nearest-neighbor algorithms) were used to harmonize synonyms, resolve spelling variations (e.g., “AI” vs. “artificial intelligence”), and consolidate semantically equivalent terms. A combination of specialized bibliometric software tools was employed to conduct performance analysis, network mapping, and thematic exploration. biblioMagika (Ahmi, 2024) served as the primary tool for computing descriptive bibliometric indicators, including yearly scientific production, citation metrics, publication growth trends, journal distribution, and lists of highly cited documents. Its preprocessing capabilities enabled the creation of a harmonized dataset for subsequent analyses. To visualize co-occurrence relationships among author keywords, VOSviewer was utilized (van Eck & Waltman, 2014). VOSviewer is a leading software for constructing and visualizing bibliometric networks, generating both network visualizations and temporal overlay maps. These outputs enabled the identification of thematic clusters and the evolution of research fronts in AI and auditing. To analyze the field’s developmental trajectory, Biblioshiny was utilized to generate the Life Cycle of Scientific Production (Aria & Cuccurullo, 2017). This tool enabled a systematic modeling of the field’s growth dynamics, providing insights into its evolutionary stage, rate of expansion, and anticipated maturation based on longitudinal publication patterns.

4. Results

The results are organized to reflect the study’s research questions. The analysis first examines the temporal evolution and life cycle of AI-in-auditing research, followed by an assessment of influential outlets and highly cited publications. It then explores the field’s conceptual structure and thematic evolution using network and overlay visualizations.

4.1. Documents profiles

The bibliometric profile of the dataset (see Table 1) reveals a maturing and increasingly influential body of scholarship on artificial intelligence in auditing.

Table 1. Citation metrics.

Main Information	Data
Publication Years	2000–2025
Total Publications	269
Citable Year	26
Number of Contributing Authors	879
Number of Cited Papers	214
Total Citations	5,049
Citation per Paper	18.77
Citation per Cited Paper	23.59
Citation per Year	201.96
Citation per Author	5.74
Author per Paper	3.27
Citation sums within h-Core	4,489
h-index	34
g-index	63
m-index	1.308

Source: Generated by the author(s) using biblioMagika® (Ahmi, 2024).

Across the 26-year publication window from 2000 to 2025, 269 journal articles were identified, collectively authored by 879 unique contributors. This output indicates sustained scholarly engagement and growing interdisciplinary collaboration, as reflected by an average of 3.27 authors per paper. Of the total publications, 214 articles had received citations, demonstrating that the majority of works have achieved scholarly recognition. The corpus accumulated 5049 citations overall, yielding an average of 18.77 citations per paper and 23.59 citations per cited paper, indicating substantial academic impact relative to the field's size. The average annual citation rate of 201.96 further underscores the field's dynamism and continued relevance.

4.2. Publication trends

The temporal progression of AI-in-auditing publications from 2000 to 2025 demonstrates a decisive evolution from isolated conceptual explorations to a robust, rapidly expanding research domain (see Table 2). The early 2000–2010 period was characterized by low publication frequency and a reliance on foundational theoretical contributions, reflecting the technological immaturity of AI tools and the profession's limited capacity to integrate automation into audit workflows. Despite

Table 2. Publication by year.

Year	TP	NCA	NCP	TC	C/P	C/CP	<i>h</i>	<i>g</i>	<i>m</i>
2000	1	3	1	1	1.00	1.00	1	1	0.038
2001	2	5	2	97	48.50	48.50	2	2	0.080
2002	3	8	2	256	85.33	128.00	2	3	0.083
2005	1	2	1	6	6.00	6.00	1	1	0.048
2006	2	4	2	14	7.00	7.00	1	2	0.050
2007	1	3	1	64	64.00	64.00	1	1	0.053
2008	2	4	2	12	6.00	6.00	2	2	0.111
2011	5	11	4	87	17.40	21.75	3	5	0.200
2012	4	5	3	14	3.50	4.67	2	3	0.143
2014	2	5	2	102	51.00	51.00	2	2	0.167
2015	1	3	1	5	5.00	5.00	1	1	0.091
2016	3	8	3	64	21.33	21.33	3	3	0.300
2017	2	6	2	372	186.00	186.00	2	2	0.222
2018	3	8	3	79	26.33	26.33	2	3	0.250
2019	5	10	5	426	85.20	85.20	5	5	0.714
2020	22	69	21	892	40.55	42.48	12	22	2.000
2021	16	40	14	371	23.19	26.50	10	16	2.000
2022	36	136	35	966	26.83	27.60	15	30	3.750
2023	31	86	29	423	13.65	14.59	12	19	4.000
2024	47	179	41	645	13.72	15.73	14	24	7.000
2025	80	284	40	153	1.91	3.83	6	9	6.000
Total	269	879	214	5049	18.77	23.59	34	63	1.308

Note: TP = total number of publications; NCA = Number of contributing authors; NCP = number of cited publications; TC = total citations; C/P = average citations per publication; C/CP = average citations per cited publication; *h* = h-index; *g* = g-index; *m* = m-index.

Source: Generated by the author(s) using biblioMagika® (Ahmi, 2024).

sparse output, several early contributions exerted substantial intellectual influence, laying the conceptual groundwork for future scholarship. Notably, Kokina and Davenport's (2017) articulation of AI's transformative implications for audit judgment and workforce structures, together with Munoko et al.'s (2020) examination of the ethical and organizational challenges surrounding AI adoption, provided conceptual anchors that have shaped how scholars frame subsequent empirical and methodological inquiries into intelligent auditing. These early contributions parallel patterns observed in other technology-intensive fields, in which seminal works provide the theoretical legitimacy and research direction that precede periods of accelerated scientific growth (Porter & Cunningham, 2004).

From 2016 onward, the publication landscape shifted markedly, propelled by significant advances in machine learning, big data infrastructures, and robotic process automation, which made AI applications more feasible and empirically demonstrable within audit environments. The period between 2020 and 2024 in particular represents an accelerated phase of scholarly activity, catalyzed in part by the COVID-19 pandemic, which heightened institutional demand for remote, automated, and data-driven audit mechanisms. Research diversified into specialized substreams, including machine-learning-based risk assessment, RPA-enabled audit procedures, and NLP-driven document analytics, supported by influential contributions such as Issa et al. (2016) and Jans et al. (2014), as well as recent developments in continuous auditing and explainable AI. The elevated h- and g-indices during these years illustrate that publications not only increased in volume but generated sustained citation traction, signaling the consolidation of AI as a core thematic pillar within auditing research. Although comparatively low citations accompany the exceptionally high publication count in 2025 due to citation lag, the trajectory aligns with broader patterns of digital transformation in the accounting literature (Appelbaum et al., 2017), indicating that AI-enabled auditing has entered a mature phase of scholarly expansion with enduring theoretical, methodological, and practical implications.

4.3. Life cycle of scientific production

The life cycle model presented in Figure 2 reveals a field experiencing sustained acceleration rather than approaching maturity. The logistic curve fitted to the annual publication data ($R^2 = 0.950$) follows the pattern of cumulative scientific growth theorized by Price (1963) and later refined in scientometric analyses that examine nonlinear, saturation-bound trajectories of research domains (Fernández-Cano et al., 2004). Consistent with empirical evidence demonstrating that scientific and technological outputs frequently follow logistic or bi-logistic growth functions (Lezama-Nicolás et al., 2018; Wong & Goh, 2010), the curve for AI-in-auditing research exhibits a steep upward trajectory across the 2019–2025 period. During these years, publication volume expanded from 5 contributions in 2019 to 80 in 2025, signaling a transition from the emergence phase to a high-growth acceleration phase.

The estimated carrying capacity ($K = 2,740$ publications) and projected midpoint ($t_m \approx 2031.5$) further suggest that the field has not yet reached its intellectual inflexion point. At the predicted peak annual output of approximately 259 publications, the model indicates that AI-in-auditing research will continue expanding well into the next decade before approaching saturation. This diffusion pattern mirrors classical innovation-adoption dynamics, in which emerging technologies experience slow initial uptake followed by rapid acceleration as conceptual legitimacy increases, methodological tools mature, and professional communities begin integrating innovations at scale (Rogers, 2003). Within auditing, these shifts are reflected in the growing incorporation of machine learning into risk assessment, the operationalization of robotic process automation (RPA), and the accelerated use of natural language processing and large language models to support evidence evaluation, developments highlighted in recent research syntheses and agenda-setting articles (Appelbaum et al., 2017; Yoon et al., 2015).

The model also provides insight into the field's expected maturation. The projected diffusion window (approximately 2025.7–2037.3) corresponds to a period in which foundational knowledge structures solidify, empirical evidence accumulates, and the profession increasingly normalizes AI-enabled audit routines. As demonstrated in studies of emergent technologies, fields typically experience a post-peak decline as innovation stabilizes, research questions narrow, and scholarly attention diffuses toward adjacent or successor technologies (Rotolo et al., 2015). In line with this pattern, the model predicts a gradual

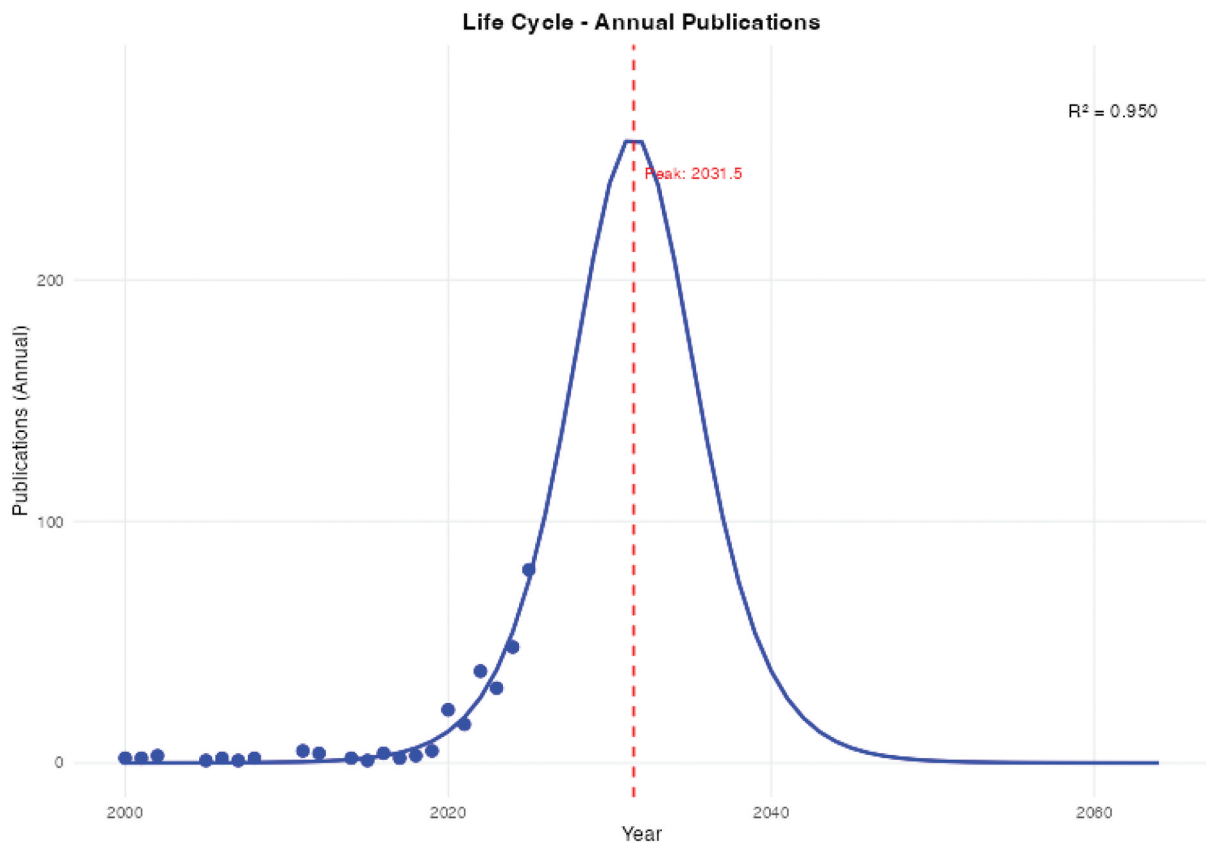


Figure 2. Life cycle of scientific production. Source: Generated by the author(s) using Biblioshiny (Aria & Cuccurullo, 2017).

reduction in publication growth after 2031 and a tapering of outputs by the 2040s, reflecting the eventual consolidation of AI-in-auditing as a mature and established research area.

4.4. Publications by journals

The distribution of publications across journals shows that AI-in-auditing research is anchored primarily in accounting information systems journals, with support from interdisciplinary technology outlets. The International Journal of Accounting Information Systems, as shown in Table 3, is the leading venue, publishing 11 papers and accumulating 610 citations, reflecting its central role in integrating intelligent

Table 3. Most active journals.

Source Title	TP	NCA	NCP	TC	C/P	C/CP	h	g	m
International Journal of Accounting Information Systems	11	32	11	610	55.45	55.45	8	11	0.333
Journal of Information Systems	8	17	8	123	15.38	15.38	7	8	0.467
Journal of Emerging Technologies in Accounting	8	17	7	504	63.00	72.00	4	8	0.444
International Journal of Data and Network Science	5	17	3	21	4.20	7.00	2	4	0.667
AI and Society	4	11	3	102	25.50	34.00	3	4	1.000
Wireless Communications and Mobile Computing	3	15	3	10	3.33	3.33	2	3	0.500
Contemporary Accounting Research	3	9	2	87	29.00	43.50	2	3	0.500
Journal of Business Ethics	3	7	2	405	135.00	202.50	2	3	0.080
Journal of Theoretical and Applied Information Technology	3	6	3	11	3.67	3.67	1	3	0.200
Intelligent Systems in Accounting, Finance and Management	3	6	2	176	58.67	88.00	2	3	0.200
Sustainability (Switzerland)	3	9	2	14	4.67	7.00	2	3	0.500
International Review of Financial Analysis	3	14	3	27	9.00	9.00	3	3	1.500
IEEE Access	3	11	2	25	8.33	12.50	2	3	0.500
Computational Intelligence and Neuroscience	3	6	3	10	3.33	3.33	3	3	0.750
Journal of Intelligent and Fuzzy Systems	3	5	2	16	5.33	8.00	1	3	0.143

Note: TP = total number of publications; NCA = Number of contributing authors; NCP = number of cited publications; TC = total citations; C/P = average citations per publication; C/CP = average citations per cited publication; h = h-index; g = g-index; m = m-index.

Source: Generated by the author(s) using biblioMagika® (Ahmi, 2024).

systems with audit practice. Similarly, the *Journal of Information Systems* and the *Journal of Emerging Technologies in Accounting* each contributed eight papers, with JETA demonstrating a substantial impact, driven by influential articles such as Kokina and Davenport's (2017) widely cited work on audit automation. These journals collectively indicate that the accounting information systems domain remains the principal platform for advancing empirical and conceptual insights into AI-enabled auditing.

Beyond these core outlets, several high-impact interdisciplinary journals also play an essential role. Titles such as *AI and Society*, *International Review of Financial Analysis*, *Journal of Business Ethics*, and *Contemporary Accounting Research*, although publishing fewer papers, exhibit disproportionately high citation-per-paper ratios. This reflects a growing interest in governance, ethics, and methodological issues surrounding AI adoption in assurance services, consistent with recent debates on algorithmic accountability and transparency in audit environments. The presence of computer science and engineering journals, such as *IEEE Access* and *Computational Intelligence and Neuroscience*, further illustrates the field's multi-disciplinary nature, in which advances in machine learning, NLP, and RPA permeate auditing research. Taken together, these publication patterns indicate that AI-in-auditing research is shaped by both accounting scholars and broader technological and ethical discourses, reinforcing the field's status as an interdisciplinary and rapidly evolving research frontier.

4.5. Highly cited documents

The citation profile, as shown in Table 4, demonstrates that a small number of influential works have disproportionately shaped the intellectual development of AI-in-auditing research. The most cited paper, Kokina and Davenport (2017), with 370 citations, introduced a conceptual framing of how automation and intelligent systems alter audit processes, decision structures, and professional roles, establishing a foundational agenda for subsequent research. Similarly, Munoko et al. (2020)'s examination of ethical implications reflects growing scholarly concern regarding algorithmic accountability, transparency, and fairness, issues now central to debates on AI governance in assurance settings. Several technically oriented works also contributed to the field's methodological grounding, including Huang and Vasarhelyi's (2019) framework for robotic process automation and Fedyk et al. (2022)'s assessment of AI-enhanced audit performance. These articles illustrate how conceptual, ethical, and technological strands have jointly advanced the field.

Table 4. Top 10 highly cited articles.

No.	Authors	Title	Source Title	Cites	Cites per Year
1	Kokina and Davenport (2017)	The emergence of artificial intelligence: How automation is changing auditing	<i>Journal of Emerging Technologies in Accounting</i>	370	41.11
2	Munoko et al. (2020)	The ethical implications of using artificial intelligence in auditing	<i>Journal of Business Ethics</i>	352	58.67
3	Huang and Vasarhelyi (2019)	Applying robotic process automation (RPA) in auditing: A framework	<i>International Journal of Accounting Information Systems</i>	217	31.00
4	Fedyk et al. (2022)	Is artificial intelligence improving the audit process?	<i>Review of Accounting Studies</i>	177	44.25
5	Rezaee et al. (2002)	Continuous auditing: Building automated auditing capability	<i>Auditing</i>	137	5.71
6	Yang et al. (2020)	Double machine learning with gradient boosting and its application to the Big N audit quality effect	<i>Journal of Econometrics</i>	137	22.83
7	Gotthardt et al. (2020)	Current state and challenges in the implementation of smart robotic process automation in accounting and auditing	<i>ACRN Journal of Finance and Risk Perspectives</i>	126	21.00
8	Calderon and Cheh (2002)	A roadmap for future neural networks research in auditing and risk assessment	<i>International Journal of Accounting Information Systems</i>	119	4.96
9	Lehner et al. (2022)	Artificial intelligence based decision-making in accounting and auditing: ethical challenges and normative thinking	<i>Accounting, Auditing and Accountability Journal</i>	104	26.00
10	Mökander et al. (2021)	Ethics-Based Auditing of Automated Decision-Making Systems: Nature, Scope, and Limitations	<i>Science and Engineering Ethics</i>	97	19.40

Source: Generated by the author(s) using biblioMagika® (Ahmi, 2024).

Other highly cited contributions underscore the field's diversification into emerging analytic techniques and specialized audit applications. Early foundational work, such as Rezaee et al. (2002) on continuous auditing and Calderon and Cheh (2002) on neural network research, demonstrates the longstanding interest in computational methods within auditing. More recent contributions, such as Yang et al. (2020) on double machine learning and Lehner et al. (2022) on ethical decision-making in AI-supported audits, highlight the field's expansion into advanced econometric modeling, automated risk assessment, and normative perspectives. These highly cited documents serve as intellectual anchors that have propelled the field from early conceptual explorations to sophisticated, interdisciplinary research addressing both the technical and societal dimensions of AI in auditing.

4.6. Author's keywords analysis

4.6.1. Network visualisation

The co-occurrence analysis examines how frequently author keywords appear together across the dataset, revealing the conceptual structure and thematic relationships within the field. Using VOSviewer (van Eck & Waltman, 2014), author keywords were mapped based on their joint appearances within the same publications, allowing the identification of dominant research themes and emerging areas of interest (see Figure 3). This technique is widely employed in bibliometric studies because keyword co-occurrence serves as a reliable proxy for the intellectual organization of a research field and helps to uncover the main knowledge clusters that underpin scholarly discourse. In this study, fractional counting was used to ensure that keywords from multi-authored papers were counted proportionally, thereby improving analytical precision. The map was generated using VOSviewer's default minimum threshold of three occurrences, which reduced the original set of 775 total keywords to 66 keywords meeting the inclusion criteria. A minimum cluster size of 7 was applied to avoid theme fragmentation and ensure that each cluster represented a substantive body

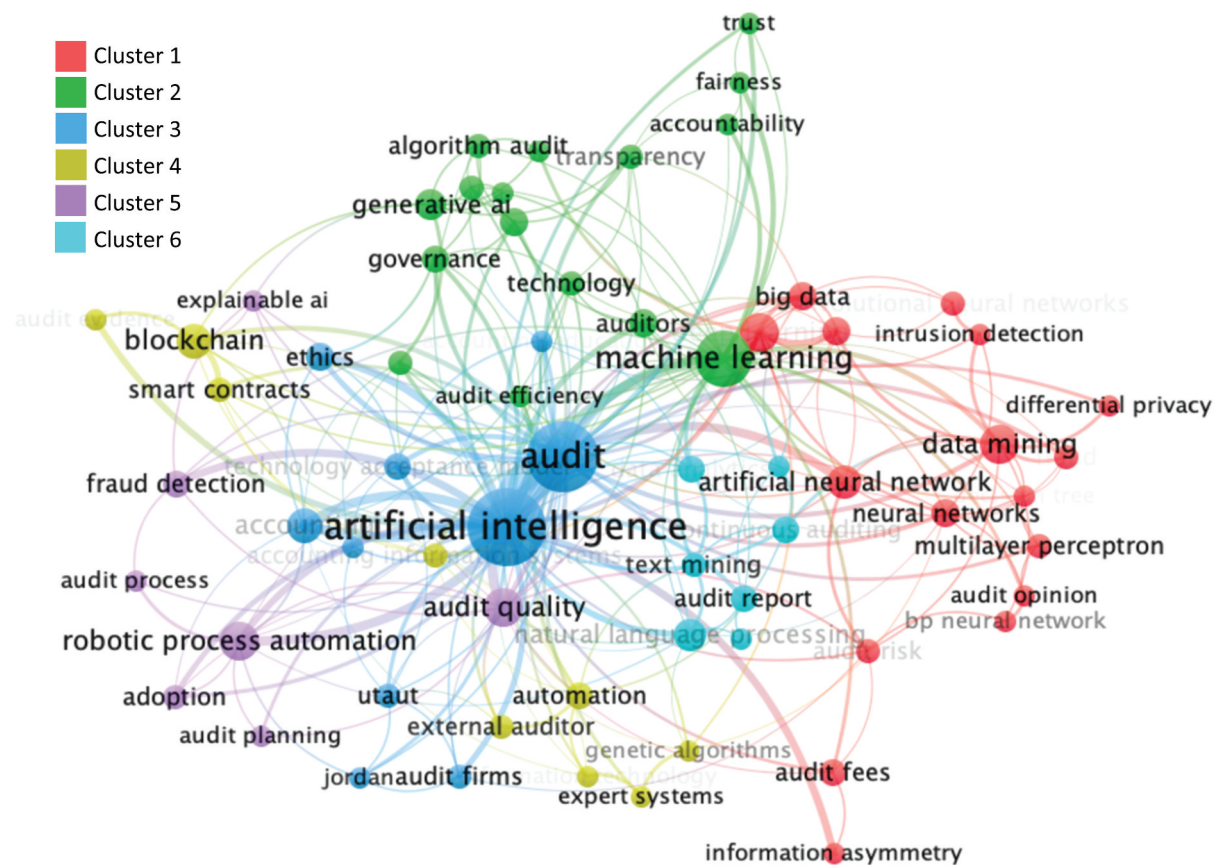


Figure 3. Network visualization of the co-occurrence analysis of the author keywords. Source: Generated by the author(s) using VOSviewer (van Eck & Waltman, 2014). View dynamic map: <https://tinyurl.com/27vgrnz9>.

of literature. Based on these settings, the software produced a map consisting of six clusters, each visualized in a distinct color. In the network visualization, the size of each node reflects the frequency of the keyword (i.e., number of occurrences). At the same time, the proximity and thickness of links represent the strength of co-occurrence relationships. Larger nodes such as “artificial intelligence,” “audit,” and “machine learning” indicate central concepts that anchor the field, while more specialized terms form satellite structures around them.

4.6.1.1. Cluster 1: Data-driven and computational approaches to audit analytics. Cluster 1 represents a technically oriented research stream in which AI and machine learning techniques are embedded directly into core audit tasks. Foundational work such as Calderon and Cheh’s (2002) roadmap for neural network research in auditing already proposed using probabilistic and multilayer architectures to support risk assessment and fraud detection, anticipating later advances. Subsequent studies broadened this line of inquiry by systematizing how different AI and analytics techniques map onto audit problems: Gray and Debreceeny (2014), for example, develop a taxonomy to guide research on the application of data mining to fraud detection in financial statement audits, thereby structuring the space in which recent deep learning work has emerged. Sun (2019) shows how deep learning can be operationalized in specific audit procedures, providing empirical evidence that neural architectures can detect anomalies in journal entries more effectively than traditional sampling approaches. Sun and Vasarhelyi (2018) extend this perspective by illustrating how textual data analytics, using data mining and machine learning techniques, enables auditors to incorporate unstructured narrative evidence into risk evaluation. Research published in *Expert Systems with Applications* further illustrates how probabilistic neural networks and other AI models can be applied to credit auditing and control tasks, demonstrating measurable improvements in classification accuracy and error reduction (Chen & Huang, 2011; Gaganis et al., 2007). Across these studies, the cluster reflects a coherent shift from conceptual proposals to increasingly sophisticated, model-driven audit analytics that leverage big data, neural networks, and deep learning to enhance fraud detection, risk scoring, and continuous anomaly surveillance.

4.6.1.2. Cluster 2: Governance, adoption, and responsible use of AI in auditing. Cluster 2 captures the socio-technical and governance dimensions of AI adoption in auditing, focusing on how organizations regulate, legitimize, and institutionalize AI-enabled tools. At the center of this cluster is work on AI’s impact on audit quality and firm behavior. Fedyk et al. (2022) provide rare large-sample archival evidence that AI investments by audit firms are associated with lower restatement risk and changes in fee structures, suggesting that AI can improve both efficiency and outcomes while simultaneously reshaping labor and pricing models. Surrounding this are normative and regulatory contributions that conceptualize what “auditing AI” entails. Mökander et al. (2021) propose ethics-based auditing frameworks for automated decision-making systems, arguing that auditability, transparency, and responsibility must be built into AI governance regimes from the outset, while their later work on conformity assessment and post-market monitoring articulates how such audits can be institutionalized in practice (Mökander et al., 2022). In the accounting and auditing domain specifically, Almaqtari (2024) links IT governance structures to the successful integration of AI and analytics in auditing, underscoring that firm-level governance mechanisms condition whether AI actually improves audit performance. More behaviorally oriented studies, such as Bin-Nashwan et al. (2025), examine how AI adoption reshapes perceptions of financial reporting and audit efficiency, drawing on technology acceptance and organizational change perspectives to show that auditor trust, perceived competence, and managerial support are critical to sustained adoption. Collectively, Cluster 2 demonstrates that AI in auditing is not merely a technical problem; it is a governance challenge involving ethics-based assurance of AI systems, organizational readiness, and the design of regulatory and institutional safeguards for algorithmic decision-making.

4.6.1.3. Cluster 3: Conceptual foundations, ethics, and behavioural contexts of AI-enabled auditing. Cluster 3 forms the conceptual and normative backbone of AI-in-auditing scholarship, integrating ethics, acceptance theories, and professional responsibility. A central strand of this cluster interrogates the ethical implications of AI-enabled audits. Munoko et al. (2020) provide a comprehensive analysis of how AI introduces new forms of bias, opacity, and gaps in responsibility in auditing,

arguing that professional standards must adapt to address issues such as algorithmic discrimination, privacy, and accountability. Lehner et al. (2022) extend this conversation by applying a structured ethics lens to AI-based decision-making in accounting and auditing, identifying tensions around objectivity, transparency, and trustworthiness and calling for new normative frameworks and ethical competencies for auditors. These contemporary concerns sit in continuity with earlier work on audit expert systems, such as Dillard and Yuthas (2001) responsibility ethic for audit expert systems, which already recognized that automation in assurance settings raises questions of moral agency and professional accountability. A second strand in this cluster involves technology acceptance and readiness. Abdullah and Almaqtari (2024) examine how AI and Industry 4.0 technologies affect accounting education and practice, employing acceptance constructs to show that perceived usefulness, attitudes toward AI, and pedagogical design shape preparedness for AI-mediated auditing. Complementing this, Anh et al. (2024) explore how technology readiness influences the adoption of AI tools in accounting and auditing, highlighting the importance of skills, infrastructure, and organizational support. Finally, the presence of education-focused work, such as Prokofieva's (2022) integration of data analytics into audit teaching, indicates recognition that conceptual, ethical, and behavioral readiness must be cultivated long before auditors engage with AI tools in practice. The cluster, therefore, anchors the field in a broader conversation about ethics, responsibility, and human capabilities in AI-enabled assurance.

4.6.1.4. Cluster 4: Blockchain, smart contracts, and technology-embedded audit evidence. Cluster 4 reflects a complementary technological ecosystem in which blockchain, smart contracts, and related infrastructures enhance the integrity, traceability, and automation of audit evidence. Within your dataset, several contributions from computer science and information systems explicitly connect blockchain architectures to auditability. Sachan and Liu (2024) focus on blockchain-based auditing of legal decisions supported by generative and explainable AI, illustrating how smart contracts and XAI can be combined to create transparent, machine-readable records that are amenable to algorithmic assurance. In an accounting-specific context, Cazazian (2022) discusses the adoption of blockchain technology in artificial-intelligence-enhanced financial systems and its implications for accounting and auditing, highlighting how decentralized ledgers and smart contracts can provide continuous, non-repudiable evidence streams that interface naturally with AI-driven analytics. Earlier strands in this cluster, such as work on expert systems and audit reporting, show how rule-based and knowledge-based systems were precursors to current combinations of blockchain, AI, and innovative infrastructures in audit tasks (e.g., Dillard & Yuthas, 2001). Together, Cluster 4 portrays a trajectory where the audit object (evidence, records, contracts) is increasingly “born digital” and technically structured, thereby enabling AI-driven assurance over system-embedded controls, transactions, and legal artifacts.

4.6.1.5. Cluster 5: Audit quality, robotic process automation, and process-level automation. Cluster 5 is centered on the practical embedding of AI and automation into audit workflows, with a strong emphasis on robotic process automation (RPA), audit quality, and process design. Huang and Vasarhelyi's (2019) framework for applying RPA in auditing illustrates how rule-based bots can perform repetitive tasks such as data extraction, reconciliations, and routine tests, thereby improving efficiency and consistency while allowing auditors to focus on higher-order analytical work. Gotthardt et al. (2020) examine the current state and challenges of innovative process automation in accounting and auditing, documenting both operational benefits and implementation barriers, including legacy systems, organizational resistance, and skills gaps. Complementing these, Zhang (2019) and Zhang et al. (2022) consider “intelligent process automation” and “attended process automation” in audit contexts, respectively, showing how human–bot collaboration can support audit planning, documentation, and evidence gathering while retaining human oversight over critical judgments. Design science work by Eulerich et al. (2022) and implementation-focused research by Perdana et al. (2023) further elaborate on frameworks and prototypes for integrating RPA into internal and external audit processes, emphasizing bot governance, performance evaluation, and integration with existing audit methodologies. This cluster shows that AI-enabled process automation is moving from conceptual proposals to implemented frameworks, raising essential questions about how automation affects audit quality, role definition, and skill requirements.

4.6.1.6. Cluster 6: Data analytics, natural language processing, and continuous auditing. Cluster 6 brings together research on advanced data analytics, natural language processing (NLP), and continuous-auditing infrastructure. Li and Liu (2020) developed an intelligent, NLP-based system for analyzing audit brainstorming sessions, illustrating how AI can capture and structure qualitative audit discussions to support risk assessment and documentation. On the assurance side, continuous auditing is a significant theme. Borthick (2012) designs continuous auditing for highly automated environments, showing how data-driven monitoring and exception reporting can be integrated into business processes and internal control systems. Raschke et al. (2018) advance this by examining AI-enhanced audit inquiry as part of continuous auditing, highlighting how AI agents can support interactive evidence gathering. Cardoni et al. (2020), in turn, link continuous auditing and data mining to strategic anti-corruption monitoring, demonstrating that continuous, analytics-driven assurance is not limited to traditional financial statement auditing but extends into broader ESG and compliance contexts. Across these studies, cloud computing and scalable data infrastructures are repeatedly emphasized as enablers of real-time analytics and continuous assurance.

4.6.2. Overlay visualisation

The overlay visualization in Figure 4 depicts the temporal progression of author keywords by average publication year, offering insight into how scholarly priorities in AI-in-auditing have evolved over the past two decades. Earlier keywords appear in green-yellow tones, while newer, cutting-edge topics emerge in orange-red hues. This gradual color transition reflects a shift from foundational AI techniques and early automation experiments toward sophisticated modeling approaches, unstructured data analytics, and ethically governed intelligent audit systems.

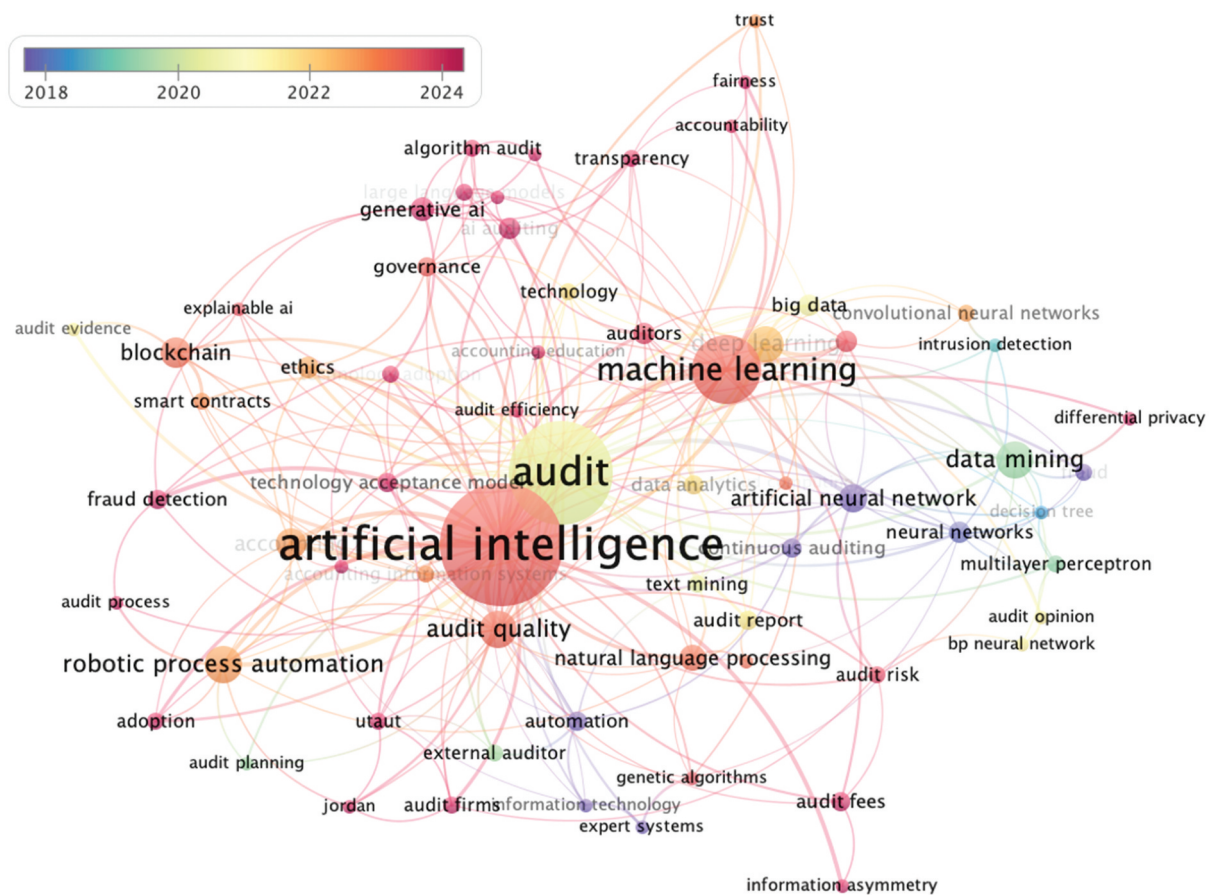


Figure 4. Overlay visualization of the co-occurrence analysis of the author keywords (Score = avg. Pub. Year). Source: Generated by the author(s) using VOSviewer (van Eck & Waltman, 2014).

4.6.2.1. Early phase: Foundations of intelligent auditing and rule-based systems. The earliest keywords, information technology, expert systems, artificial neural networks, and automation, reflect a formative period in which research primarily explored whether traditional AI and expert systems could support audit judgment and structured decision-making. During this stage, scholars built upon early work such as Calderon and Cheh's (2002) neural-network roadmap for auditing and Dillard and Yuthas's (2001) ethical considerations surrounding audit expert systems, which set the conceptual foundation for later machine-learning-based audit research. Information systems research from this era frequently examined how rule-based automation, IT controls, and knowledge-based systems could improve audit documentation, evidence gathering, or risk signaling, anticipating later developments in continuous auditing and analytics-driven assurance.

4.6.2.2. Expansion phase: Data-driven and analytics-enhanced auditing. From 2017 onward, the field shifted markedly toward machine learning, big data, and continuous auditing applications. Keywords such as fraud, continuous auditing, neural networks, decision trees, and data mining signal the rise of analytics-driven research that exploits increasingly rich audit datasets. This aligns with emerging scholarly contributions that leverage big-data infrastructures to produce full-population testing and anomaly detection in financial records, as seen in Sun's (2019) applications of deep learning to journal entry testing and Sun and Vasarhelyi's (2018) application of textual analytics to audit evidence processing. Simultaneously, data-centric audit processes became more integrated into core audit tasks. The appearance of keywords such as audit planning, audit opinion, big data, and text mining suggests that auditors were increasingly incorporating predictive models and unstructured data analytics into substantive testing and risk assessment. These developments parallel the maturing continuous-auditing frameworks described by Borthick (2012) and the emergence of AI-enhanced audit inquiry tools documented by Raschke et al. (2018). During this period, the field transitioned from conceptual curiosity toward operational integration of machine-learning models.

4.6.2.3. Acceleration phase: Advanced architectures, RPA, and ethical-governance concerns. Beginning around 2022, the overlay visualization reveals a surge of sophisticated AI techniques and heightened attention to governance. Keywords such as deep learning, convolutional neural networks, natural language processing, and robotic process automation illustrate a move toward more powerful model architectures capable of analyzing unstructured financial data, automating multi-step audit workflows, and generating richer audit insights. This trend aligns with the widespread adoption of RPA frameworks in auditing, as outlined by Huang and Vasarhelyi (2019), and with operational implementations, such as those outlined by Li and Liu (2020). Parallel to this technical sophistication, normative and socio-technical concerns became more prominent. Keywords such as ethics, trust, transparency, governance, and fairness echo the heightened scrutiny over AI governance and responsible use, reflected in influential research by Munoko et al. (2020), Lehner et al. (2022), and Mökander et al. (2021, 2022), who collectively argue for auditability, explainability, and ethical oversight of automated decision systems. The overlay pattern suggests that the field began to divide along two parallel trajectories: (1) technical expansion into sophisticated predictive and generative models, and (2) regulatory, behavioral, and ethical structures necessary to ensure responsible adoption within audit practice.

4.6.2.4. Emerging frontier: Generative AI, explainability, and algorithmic oversight. The most recent research frontier is defined by intensely red-shaded keywords such as explainable AI, generative AI, large language models, computer vision, and differential privacy. These terms represent the cutting edge of AI-in-auditing research, characterized by transformative technologies that are still in early adoption but showing rapid momentum. Generative AI and LLMs signal a shift toward models capable of producing narrative insights, summarizing audit evidence, and conducting semantic analysis, capabilities that extend well beyond earlier rule-based or numeric classifiers. Their emergence aligns with works such as Sachan and Liu (2024), who explore generative AI in legally binding decision-making audits, and Anh et al. (2024), who examine AI adoption readiness in financial and audit contexts. At the same time, oversight-oriented keywords such as algorithm audit, audit efficiency, and technology acceptance model indicate a growing need to evaluate, rather than deploy, AI systems in auditing. This aligns with Fedyk et al. (2022), who empirically assess AI's impact on audit quality, and with behavioral studies such as Bin-Nashwan et al.

(2025) that examine how auditors perceive AI's influence on reporting reliability and professional judgment. Privacy-preserving techniques such as differential privacy demonstrate that researchers are now addressing data-governance constraints inherent in the use of large, sensitive financial datasets in algorithmic audit systems. Together, these emerging topics indicate that AI-in-auditing research is entering a new era marked by sophisticated generative models, explainability requirements, and advanced privacy-protection mechanisms, all poised to redefine audit evidence, audit judgment, and the structure of assurance engagements.

5. Discussion

This study provides a comprehensive mapping of how AI-in-auditing research has developed over 26 years, revealing a field undergoing rapid expansion and increasing conceptual sophistication. Publication activity accelerates sharply after 2019, coinciding with the profession's transition toward data-centric audit methodologies and the emergence of advanced AI applications. The citation landscape highlights several foundational contributions spanning computational modeling, behavioral auditing under automation, and the ethical and governance implications of AI-enabled decision-making. The results show that six interrelated clusters constitute the field's intellectual structure: computational audit analytics; governance and responsible AI; foundational AI-audit integration; blockchain-based audit evidence; automation and audit quality; and analytics-driven continuous auditing. Temporal patterns illustrate a clear shift from early deterministic systems toward explainable AI, deep learning, and generative AI, reflecting the growing relevance of judgment, governance, and assurance concerns as AI becomes embedded in audit contexts.

The thematic evolution uncovered in this study mirrors broader transformations in organizational technology environments and audit practice. Early research aligned with rule-based audit logic and expert systems, whereas the subsequent proliferation of enterprise systems and big-data infrastructures enabled the development of machine-learning-driven audit analytics capable of processing complex, unstructured information. This transition reshapes traditional conceptions of audit evidence, risk assessment, and materiality. The recent prominence of governance-related concepts, ethics, transparency, trust, and fairness signals recognition that AI adoption extends beyond technical implementation to questions of accountability, explainability, and professional responsibility. Simultaneously, the emergence of generative AI and large language models introduces new uncertainties regarding cognitive reliance, documentation integrity, and judgment quality. These findings suggest a shift from viewing AI as a tool that enhances audit efficiency to a socio-technical system that materially influences auditor behavior, organizational processes, and the broader assurance ecosystem.

5.1. Implications for theory and practice

The results have important implications for the theoretical foundations of auditing. Traditional judgment frameworks must account for the cognitive consequences of interacting with opaque or probabilistic models, including how AI influences skepticism, perceptions of evidence reliability, and decision strategies. Governance-focused findings highlight the need for theory-driven models that integrate explainability, accountability, and algorithmic oversight into the audit process. Furthermore, emerging generative technologies call into question existing assumptions regarding audit evidence, documentation, and the division of labor between human auditors and intelligent systems. In practice, audit firms must balance the efficiency gains from automation with the risks of over-reliance, misinterpretation of AI-generated insights, and challenges related to data governance and privacy. Regulators may draw on these findings to refine standards addressing model transparency, testing protocols, and auditor responsibilities in AI-mediated environments.

5.2. Recommendations for future research

Future research should move beyond documenting technological uptake to theorizing and testing how AI reshapes audit work as a socio-technical practice. A priority concerns auditors' behavioral and cognitive responses to AI-enabled evidence and recommendations. Prior behavioral auditing research indicates that

advanced analytics can reconfigure auditors' information processing and may heighten risks of automation bias, over-reliance, and weakened skepticism under certain conditions (Brown-Liburd et al., 2015). What remains under-examined is how auditors interrogate, contest, and justify algorithmic outputs when tasks are ill-structured or judgment-laden (e.g., evaluating management estimates, anomalies, and narrative inconsistencies). Building on emerging audit-analytics applications, such as AI-enhanced inquiry (Raschke et al., 2018) and NLP-supported audit planning knowledge discovery (Li & Liu, 2020), future studies should use experiments, process-tracing, and longitudinal field designs to observe calibration, escalation pathways (override vs. acceptance), and team-level sensemaking when humans and models disagree.

A second avenue concerns governance and accountability mechanisms for AI within audit engagements and audit firms. The growing prominence of ethics, transparency, and algorithm auditing in the field signals that assuring the AI is becoming inseparable from assuring the financial statements (Lehner et al., 2022; Munoko et al., 2020). However, empirical evidence remains limited regarding how audit organizations operationalize responsible-AI principles through model validation, documentation, monitoring, and assignment of accountability across auditors, data specialists, and vendors. This gap is increasingly consequential as regulatory conversations about conformity assessment and post-market monitoring explicitly position "auditing" as a governance instrument for AI systems (Mökander et al., 2021, 2022). Accordingly, future research should examine which governance designs are feasible under audit constraints (confidentiality, independence, time budgets), how "explainability" is evaluated as audit evidence, and which assurance artifacts (model cards, audit trails, control testing) are most persuasive to reviewers, regulators, and audit committees.

Third, relevant scholarship can be advanced by treating AI diffusion as institutionally conditioned rather than technologically deterministic. Evidence from practice-oriented contexts shows that external shocks and professional pressures can accelerate digital transformation in auditing: during COVID-19, remote audit technologies were rapidly adopted, especially under resource constraints and normative pressures to sustain audit continuity (Sharma et al., 2022). Complementing this, large-scale evidence suggests that AI adoption can materially affect audit outcomes, shaping efficiency indicators and audit-quality proxies, while raising questions about the organizational conditions under which such gains are realized (Rahman et al., 2024). These findings motivate cross-country and cross-firm comparative studies grounded in institutional theory and organizational change: why do some jurisdictions and firms institutionalize AI faster, which configurations of digital infrastructure and governance enable sustainable use, and when does adoption reallocate (rather than reduce) audit effort toward higher-risk areas?

Finally, methodological innovation should align AI evaluation with auditing's assurance objectives. As implementation studies continue to show that automating audit work involves socio-technical trade-offs and organizational redesign (e.g., RPA prototyping, governance arrangements, and capability development) (Perdana et al., 2023), future work should develop audit-specific evaluation frameworks that connect model performance to materiality, risk thresholds, and evidence sufficiency rather than generic predictive accuracy. This includes (i) multimodal audit analytics that integrate financial, textual, process, and network data; (ii) hybrid approaches that combine rule-based audit logic with learning systems to improve interpretability; and (iii) privacy-preserving techniques suitable for audit environments, consistent with the field's emerging attention to data governance and confidentiality constraints. Together, these directions would strengthen both theory-building and practical relevance by explaining not only whether AI changes auditing, but through which behavioral, organizational, and governance mechanisms those changes translate into audit quality, legitimacy, and trust.

6. Conclusion

This study provides the first comprehensive, longitudinal science-mapping analysis of AI-in-auditing research spanning 26 years. The findings reveal a field undergoing rapid transformation, characterized by escalating publication activity, increasing thematic complexity, and growing attention to ethical, behavioral, and governance concerns. By identifying six interconnected thematic clusters and tracing their chronological progression, the study illuminates the intellectual architecture that underpins contemporary scholarship. The life-cycle modeling further demonstrates that

AI-in-auditing research remains in an expansionary stage, suggesting that conceptual innovation, methodological diversification, and practical relevance will continue to intensify in the coming decade.

Beyond synthesizing the existing knowledge base, the study provides a theoretically meaningful agenda for future research. Key opportunities lie in understanding how auditors interact cognitively and behaviorally with intelligent systems, how organizations and regulators design governance structures for AI-enabled assurance, and how emerging technologies such as large language models and explainable AI reshape evidence collection, judgment processes, and professional responsibilities. The findings also emphasize the value of cross-country and institutional comparisons, as well as the need for audit-specific methodological innovations that align AI capabilities with the core principles of independence, skepticism, and public trust. By offering a consolidated, empirically grounded view of how AI-in-auditing research has developed, this study provides scholars, practitioners, and regulators with a clearer understanding of the field's trajectory and its future possibilities. As the audit profession stands at the threshold of profound technological change, the insights generated here serve not only to map past progress but also to inform theory-building, methodological advancement, and responsible innovation in the era of intelligent auditing.

Disclosure statement

No potential conflict of interest was reported by the author(s).

ORCID

Aidi Ahmi  <http://orcid.org/0000-0002-8488-6966>

Louis Smidt  <http://orcid.org/0000-0003-0362-1075>

Declaration of generative AI and AI-assisted technologies

During the preparation of this work, the author(s) used ChatGPT to do the paraphrasing and some data interpretation. After using this tool/service, the author(s) reviewed and edited the content as needed and take full responsibility for the content of the publication.

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